

ECE 4502/6502 & CS 6501 Machine Learning on Graphs: Homework 1

Due: 02/19/2025 23:59 PM EST

Note: The assignment must be submitted electronically on Canvas.

Problems

Problem 1: Adjacency Matrix (25 pts)

Consider an undirected graph G with adjacency matrix A . Denote A_i as the i th row and A_{ij} as the element at row i and column j .

- How can we compute the number of common neighbors between vertex i and vertex j ? (3 pts)
- The number of triangles in the graph G ? (5 pts)
- The number of walks of length k ? (5 pts)
- What is challenging about computing the number of paths using the adjacency matrix? (6 pts)
- How can we determine if the graph is connected using the adjacency matrix? (6 pts)

Problem 2: Shortest Paths and Minimum Spanning Tree (25 pts)

1. Detail why edges with negative weights are not desirable for computing shortest paths. (6 pts)

2. Compute the shortest path between v_1 and v_9 using Dijkstra's algorithm for the graph in Figure 1. Provide step-by-step details. (8 pts)
3. Find a minimum spanning tree for the graph in Figure 1. Provide step-by-step details. (11 pts)

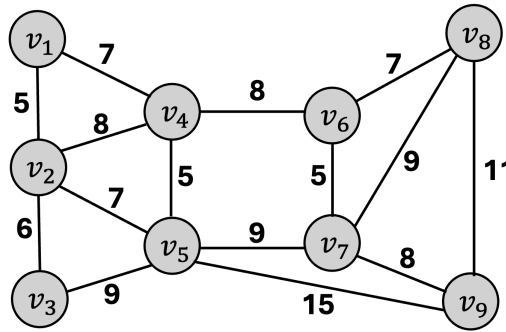


Figure 1: The input graph for Problem 2.

Problem 3: PageRank (25 pts)

1. Calculate PageRank values for the graph in Figure 2 when:
 - $\alpha = 0.3, \beta = 0.5$.
 - $\alpha = 0, \beta = 1$. (6 pts)
2. Discuss the effects of different values of α and β for this particular problem. (4 pts)
3. Does β have any effect on the order of centralities? If for one value of β the centrality value of node v_i is greater than that of v_j , is it possible to change β in a way such that v_j 's centrality becomes larger than v_i 's? (5 pts)
4. In PageRank, what α values can we select to guarantee that centrality values are calculated correctly (i.e., values do not diverge)? (5 pts)
5. Prove that the largest eigenvalue of matrix AD^{-1} is 1. (5 pts)

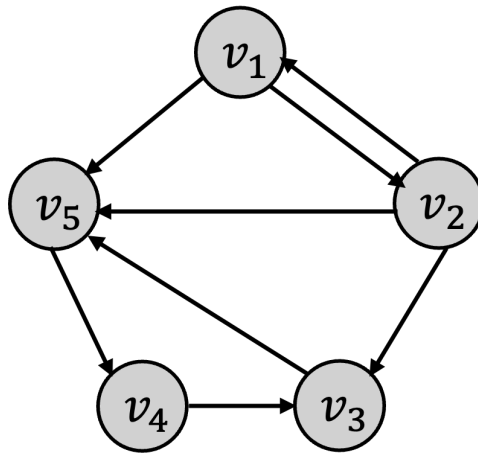


Figure 2: The input graph for Problem 3.

Problem 4: Graph Analysis (25 pts)

1. Import the Les Miserables graph (the coappearance network of characters in the novel *Les Miserables*) and visualize the graph. (3 pts)
2. Node centrality: Calculate the degree centrality, eigenvector centrality, Katz centrality, PageRank centrality, betweenness centrality, and closeness centrality for the nodes in the graph. List the top 5 nodes and their corresponding centrality scores for each centrality measure. (7 pts)
3. Graph measures: Calculate and report the degree distribution, global clustering coefficient, and diameter for the graph. (5 pts)
4. Implement PageRank centrality with power method. Start the power method with uniform importance scores, and report the iteration step needed to converge (i.e., report t such that $\|c_t - c_{t-1}\|_2 \leq 0.01$). (10 pts)